

An eight-pole, 25-kW, 130-V DC generator has a duplex lap-wound armature, which has 64 coils with 16 turns per coil. Its rated speed is 2400 r/min. Find the current per path, if $a = 16$ parallel current paths through the machine.

Select one:

- ☐ a. 11.2 A
- ☐ b. 10.4 A
- ☒ c. 12 A
- ☐ d. 9.2 A
- ☐ e. 9.8 A



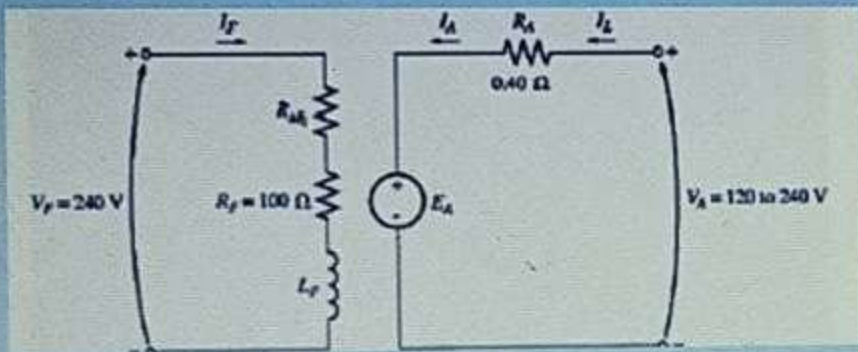
Question 23

Not yet answered

Marked out of 2.00

Flag question

For the DC motor in the figure, the generated voltage E_{AO} of 255 V at speed of 1250 PRM, what is the speed if the new terminal voltage is 130 V.



Select one:

- ☐ a. 539 RPM
- ☐ b. 686 RPM
- ☐ c. 490 RPM
- ☒ d. 637 RPM
- ☐ e. 588 RPM

Question 22

Not yet answered

Marked out of 2.00

Flag question

The angular velocity of a motor shaft rotating at 2100 rpm is

Select one:

- ☒ a. 219.9 rad/s
- ☐ b. 209.4 rad/s
- ☐ c. 230.4 rad/s
- ☐ d. 251.3 rad/s
- ☐ e. 240.9 rad/s



5	6	7	8
3	14	15	16
1	22	23	24

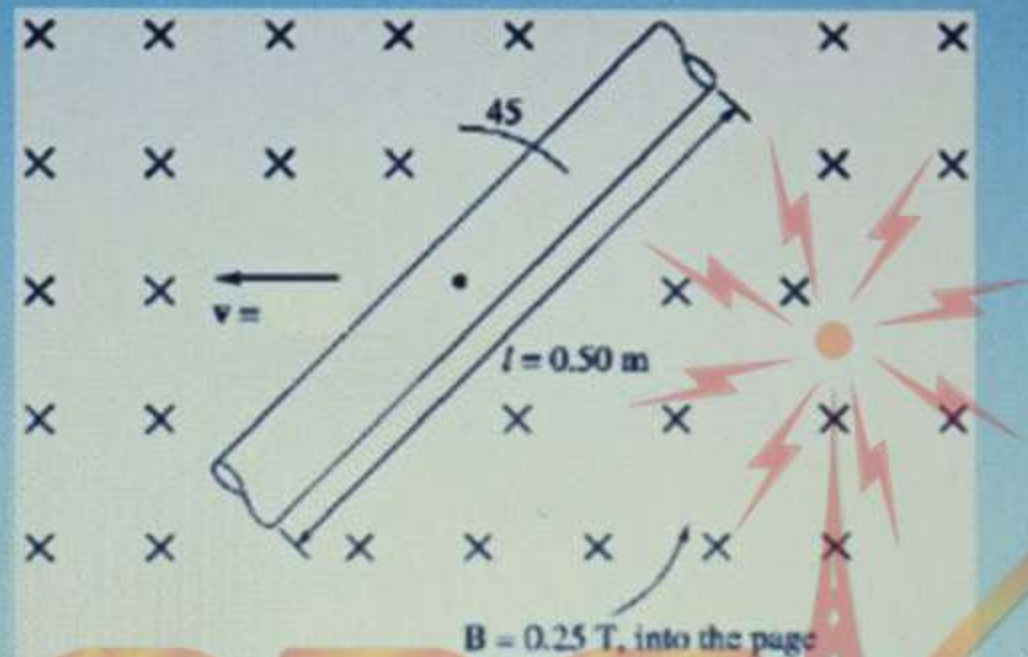
Question 19

Not yet answered

Marked out of 2.00

Flag question

Calculate the induced voltage in the wire in figure below if $v = 24 \text{ m/s}$.



Select one:

- ☐ a. 2.03 V
- ☒ b. 2.12 V
- ☐ c. 1.94 V
- ☐ d. 2.30 V
- ☐ e. 1.86 V

Question 18

Not yet answered

Marked out of 2.00

Flag question

What is the primary full load current of a 20 kVA 2500/200 V single phase transformer, the units in Amperes

Select one:

- ☐ a. 20
- ☐ b. 16
- ☐ c. 10
- ☐ d. 32
- ☒ e. 8

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Question 16

Not yet answered

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Flag question

For a shunt DC motor, the load current is 190 A, terminal voltage is 240 V, total resistance $R_F + R_{adj}$ of 40 Ω , and armature resistance of 0.09 Ω , Find generated voltage E_A .

Select one:

- ☐ a. 217.2 V
- ☐ b. 219.1 V
- ☐ c. 226.7V
- ☒ d. 222.9V
- ☐ e. 224.8 V

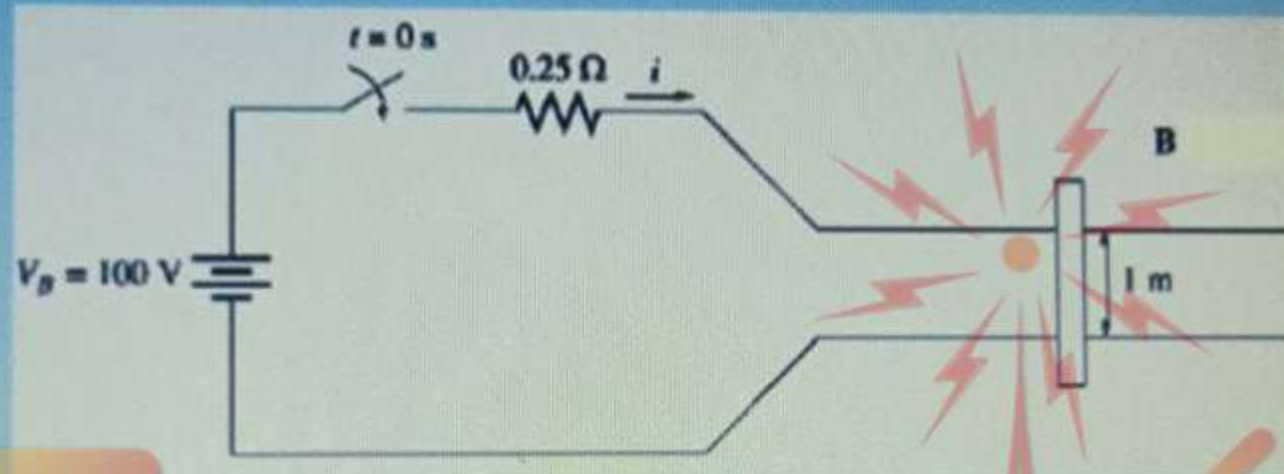
Question 24

Not yet answered

Marked out of 2.00

Flag question

Calculate the no-load steady-state speed of the bar in the figure if magnetic density $B=1.9\text{ T}$.



Select one:

- ☐ a. 58.8 m/s
- ☐ b. 47.6 m/s
- ☐ c. 50.0 m/s
- ☐ d. 55.6 m/s
- ☒ e. 52.6 m/s

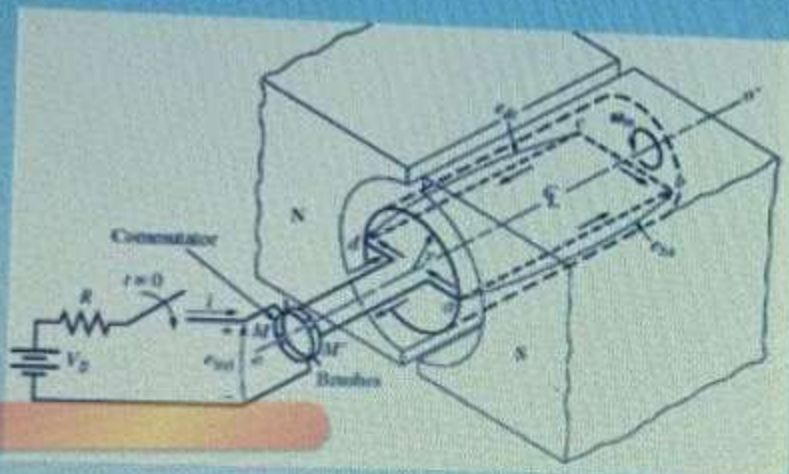
Question 14

Not yet answered

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Flag question

Given rotor speed $\omega = 480$ in the figure below, magnetic flux $B = 0.25$ T, conductor radius $r = 0.5$ m, and conductor length $l = 1$ m. Find the induced voltage.



Select one:

- ☐ a. 120 V
- ☐ b. 140 V
- ☐ c. 160 V
- ☐ d. 130 V
- ☐ e. 150 V

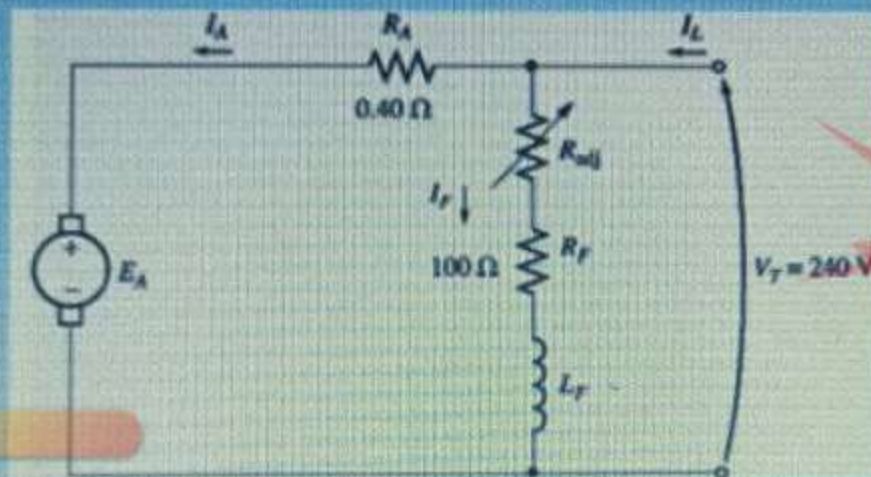
Question 15

Not yet answered

Marked out of 2.00

Flag question

The DC shunt motor shown in the figure has the resistor R_{adj} is equal to $380\ \Omega$, find the field current.



Select one:

- ☐ a. 0.67A
- ☐ b. 0.63A
- ☒ c. 0.5A
- ☐ d. 0.57A
- ☐ e. 0.65A

n 13

answered

out of 2.00

question

12 pole dc generator has a simplex wave-wound armature containing 144 coils of 10 turns each. The resistance of each turns is 0.011Ω . Its flux per pole is 0.16 Wb and it is turning at speed of 200 rpm . What is the induced countertorque on the machine shaft if load terminal is 1000Ω .

Select one:

- ☐ a. 1267 N.m
- ☐ b. 1109 N.m
- ☐ c. 1030 N.m
- ☐ d. 1188 N.m
- ☐ e. 950 N.m



Question 12

Not yet answered

Marked out of 2.00

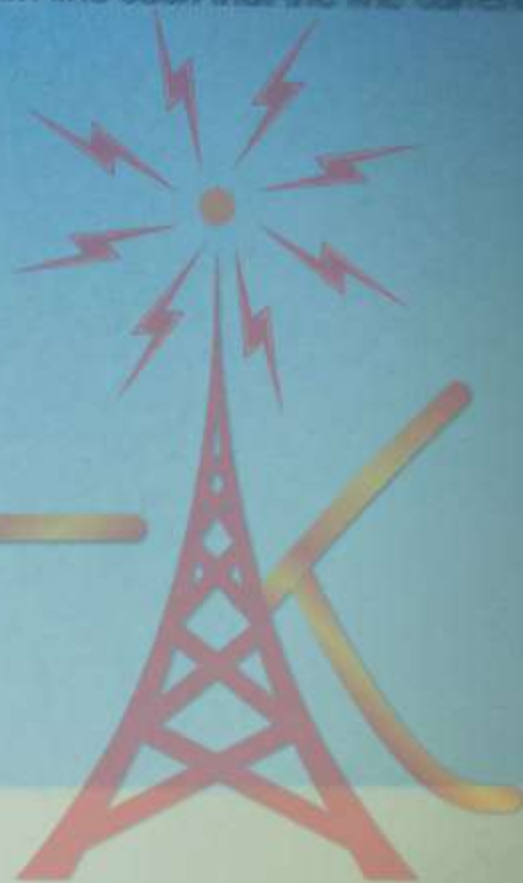
Flag question

A 480 V generator is supplying a load through a transmission line such that the line current is $10 \angle 45^\circ$ A. Find the line losses of $Z_{\text{line}} = 0.6 + j2 \Omega$.

Select one:

- ☐ a. 80 W
- ☐ b. 60 W
- ☐ c. 90 W
- ☐ d. 70 W
- ☐ e. 100 W

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Question 10

Not answered

1 out of 2.00

g question

Find the mechanical power a motor delivered to a load, if a motor is supplying 100 N.m and motor shaft is rotating at 2000 rpm.

Select one:

- ☐ a. 25.13 kW
- ☐ b. 21.99 kW
- ☐ c. 24.09 kW
- ☐ d. 20.94 kW
- ☐ e. 23.04 kW

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Question 11

Not answered

Scored out of 2.00

Flag question

Assume a DC motor has 100% efficiency and running at speed of 1000 rpm, what is the torque if E_A voltage is 6 Volts, and armature current is 150 mA.

Select one:

- ☐ a. 9.17 mN.m
- ☐ b. 8.59 mN.m
- ☐ c. 6.88 mN.m
- ☐ d. 8.02 mN.m
- ☐ e. 7.45 mN.m

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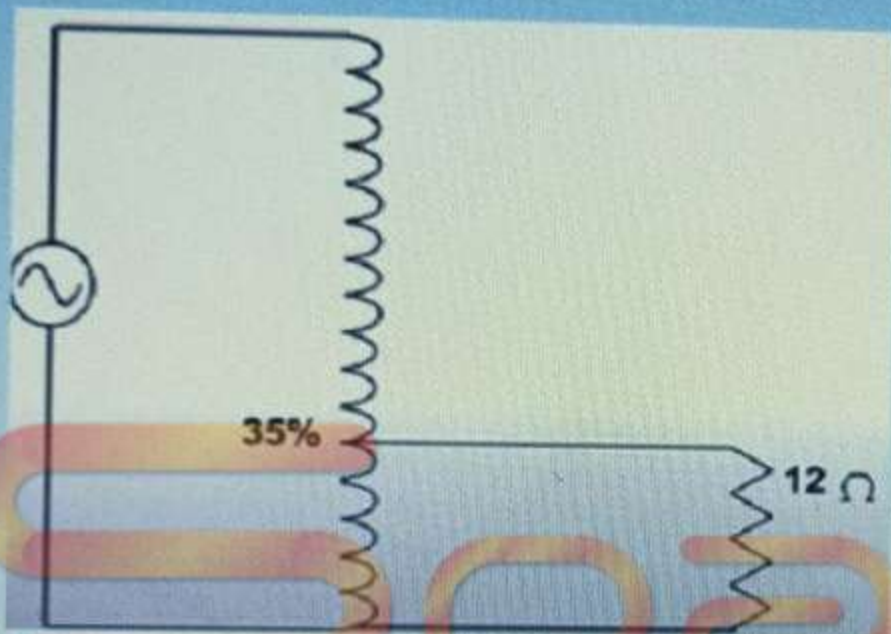
Question 17

Not yet answered

Marked out of 2.00

Flag question

A step down autotransformer shown in the figure below, find current in the $12\ \Omega$ load if the supply voltage is 60 V .



Select one:

- ☐ a. 2.33 A
- ☒ b. 1.75 A
- ☐ c. 1.46 A
- ☐ d. 2.04 A
- ☐ e. 2.63 A

Question 9

Not yet answered

Marked out of 2.00

Flag question

Given voltage $V_B = 70 \text{ V}$ in the figure below, magnetic flux $B = 0.25 \text{ T}$, conductor radius $r = 0.5 \text{ m}$, and conductor length $l = 1 \text{ m}$. Find the rotor speed ω .

Select one:

- ☐ a. 400 rad/s
- ☐ b. 280 rad/s
- ☐ c. 360 rad/s
- ☐ d. 320 rad/s
- ☐ e. 240 rad/s

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Question 8

Not yet answered

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A duplex lap-wound armature is used in a six-pole dc machine with six brush sets, each spanning two commutator segments. There are 72 coils on the armature, each containing 12 turns. The flux per pole in the machine is 0.039 Wb, and the machine speed is 240 rpm. Find the induced voltage E_A .

Select one:

- ☐ a. 112.3 V
- ☐ b. 123.6 V
- ☐ c. 157.2 V
- ☐ d. 134.8 V
- ☐ e. 146 V

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Question 5

Not yet answered

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Flag question

For a shunt DC motor without compensating winding, terminal voltage is 240 V, total resistance $R_F + R_{adj}$ of 80 Ω , and armature resistance of 0.07 Ω . There are 1200 turns per pole produces demagnetization mmf of 1200 A.turns at load current of 190 A. Find effective shunt current.

Select one:

- ☐ a. 11 A
- ☐ b. 3 A
- ☐ c. 1 A
- ☐ d. 5 A
- ☐ e. 2 A

Question 6

Not yet answered

Marked out of 2.00

Flag question

Given a 18 kVA 1200/120 V Δ -Y transformer, find primary rated current.

Select one:

- ☐ a. 5.27 A
- ☒ b. 4.44 A
- ☐ c. 4.72 A
- ☐ d. 5 A
- ☐ e. 4.17 A

POLYTECHNIC

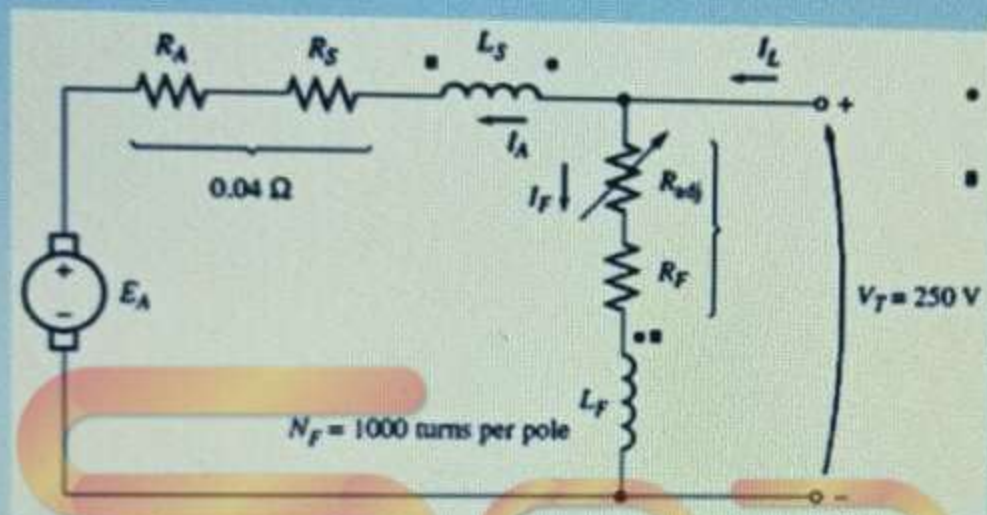
Question 4

Not yet answered

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Flag question

The cumulatively compounded DC motor without compensating winding is shown in the figure, the 1000 turns per pole produces demagnetization mmf of 1300 A.turns. The series windings is 10 turns and the armature current is 190A. Given the field current of 5 A will be produced at 250 at 1200 rpm, find the effective shunt current.



Select one:

- ☐ a. 5.6 A
- ☐ b. 6 A
- ☐ c. 5.9 A
- ☐ d. 5.8 A
- ☐ e. 5.7 A

Question 7

Not yet answered

Marked out of 2.00

Flag question

An eight-pole, 25-kW, 170-V DC generator has a duplex lap-wound armature, which has 64 coils with 16 turns per coil. Its rated speed is 2400 r/min. Find the flux per pole to produce the rated voltage in this generator at no-load conditions

Select one:

- ☐ a. 2.93 mWb
- ☐ b. 3.91 mWb
- ☐ c. 3.42 mWb
- ☐ d. 3.17 mWb
- ☐ e. 4.15 mWb

3	4	5	6	7	8
11	12	13	14	15	16
19	20	21	22	23	24

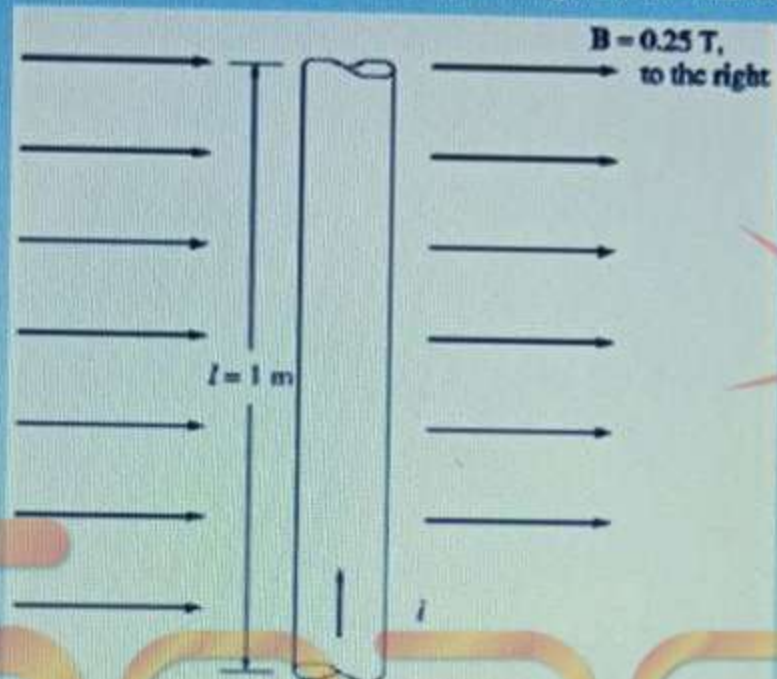
Question 2

Not yet answered

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Flag question

Calculate the force on the wire shown in figure if the wire is carrying 21 A.



Select one:

- ☐ a. 5.5 N
- ☐ b. 5.25 N
- ☐ c. 5.75 N
- ☐ d. 6 N
- ☐ e. 5 N

Question 1

Not yet answered

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Flag question

A single phase transformer with 20 kVA, 8000/240 V, has the following open circuit parameters, $V_{OC} = 8000$ V, $I_{OC} = 0.18$ A, and $P_{OC} = 400$ W. Find the open power factor for open-circuit test.

Select one:

- ☐ a. 0.46
- ☐ b. 0.5
- ☐ c. 0.42
- ☐ d. 0.28
- ☐ e. 0.33